

Comprehensive Analysis and Dietary Strategies with Tableau

A College Food Choices Case Study • Folder 4: Data Preprocessing Phase

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Submission Date:	July 2026

Folder 4 Metadata & Preprocessing Registry

Registry Parameter	Value Specification
Repository Phase	Folder 4 – Data Preprocessing (Cleaning & Transformation)
Primary Objective	Standardize Raw Survey Inputs, Handle Missing/Null Values, Type Casting
Processing Tool	Microsoft Excel / Tableau Data Interpreter Engine

1. Data Cleaning Executive Summary

Preprocessing is the critical stage where raw, unstructured survey logs are translated into machine-readable datasets. This phase involves removing incomplete entries, standardizing categorical variables (e.g., gender, budget tiers), and ensuring numeric fields are correctly cast for computational logic in the final Tableau visualization environment.

2. Preprocessing Logic Steps

Operation ID	Data Field	Transformation Logic
CLN-01	Missing Records	Rows with > 40% null responses are excluded to maintain aggregate accuracy.
CLN-02	Categorical Mapping	Gender strings converted to standardized integer indices (1/2) for analysis.
CLN-03	Numeric Casting	GPA and calorie strings forced into float/integer types for mathematical aggregation.

3. Data Handling Validation

Following the initial cleaning, the dataset was validated against the schema defined in Folder 2. Key quality assurance checks included:

- **Range Verification:** Ensuring GPA values remain within the valid [0.0 - 4.0] constraint.
- **Outlier Sanitization:** Removing extreme budget inputs that fall outside three standard deviations.
- **Consistency Checks:** Ensuring binary choice variables contain only the assigned numerical values.

4. Folder 4 Conclusion

The preprocessing phase is complete. The dataset is now sanitized, indexed, and ready for ingestion into the final Tableau engine. This clean state ensures that dashboard calculations (such as Dietary Quality Index and Stress-Diet correlations) will execute accurately without data distortion.

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